

Airkrit India Pvt. Ltd.

D-160, Dwarka, New Delhi – 110075

CIN - U85499DL2023PTC423549

www.airkrit.com

support@airkrit.com

+91 8929252575



AIRKRIT – EDUCATION to EMPLOYMENT

Where Learning Meets Real-World Execution

Problem Statement

About the Data File

This dataset contains insights into credit card transactions made in India, offering a comprehensive look at the spending habits of Indians across the nation.

- City: The city in which the transaction took place. (String)
- Date: The date of the transaction. (Date)
- Card Type: The type of credit card used for the transaction. (String)
- Exp Type: The type of expense associated with the transaction. (String)
- Gender: The gender of the cardholder. (String)
- Amount: The amount of the transaction. (Number)

These topics you will learn when you solve this case study

- Date functions: DATEDIFF, DATEPART, DATEADD
- CASE WHEN Statement
- Aggregate functions with CASE WHEN, GROUP BY, ORDER BY
- Joins
- Window functions (DENSE_RANK, LEAD, LAG)
- CTE (common table expression)

Tasks

1. Write a query to print top 5 cities with highest spends and their percentage contribution of total credit card spends.
2. Write a query to print highest spend month and amount spent in that month for each card type.
3. Write a query to print the transaction details (all columns from the table) for each card type when it reaches a cumulative of 1000000 total spends (We should have 4 rows in the o/p one for each card type).
4. Write a query to find city which had lowest percentage spend for gold card type.
5. Write a query to print 3 columns: city, highest_expense_type, lowest_expense_type (example format : Delhi, bills, Fuel).
6. Write a query to find percentage contribution of spends by females for each expense type.

7. Which card and expense type combination saw highest month over month growth in Jan-2014.
8. During weekends which city has highest total spend to total no of transaction's ratio?
9. Which city took least number of days to reach its 500th transaction after first transaction in that city?

Programming Language and Tools

- SAS, Python

Linkedin : <https://www.linkedin.com/company/airkritindia/>

Instagram : https://www.instagram.com/airkrit_india/

Thanks and Regards

